

The Brainstem

(Medulla oblongata, Pons & Midbrain)



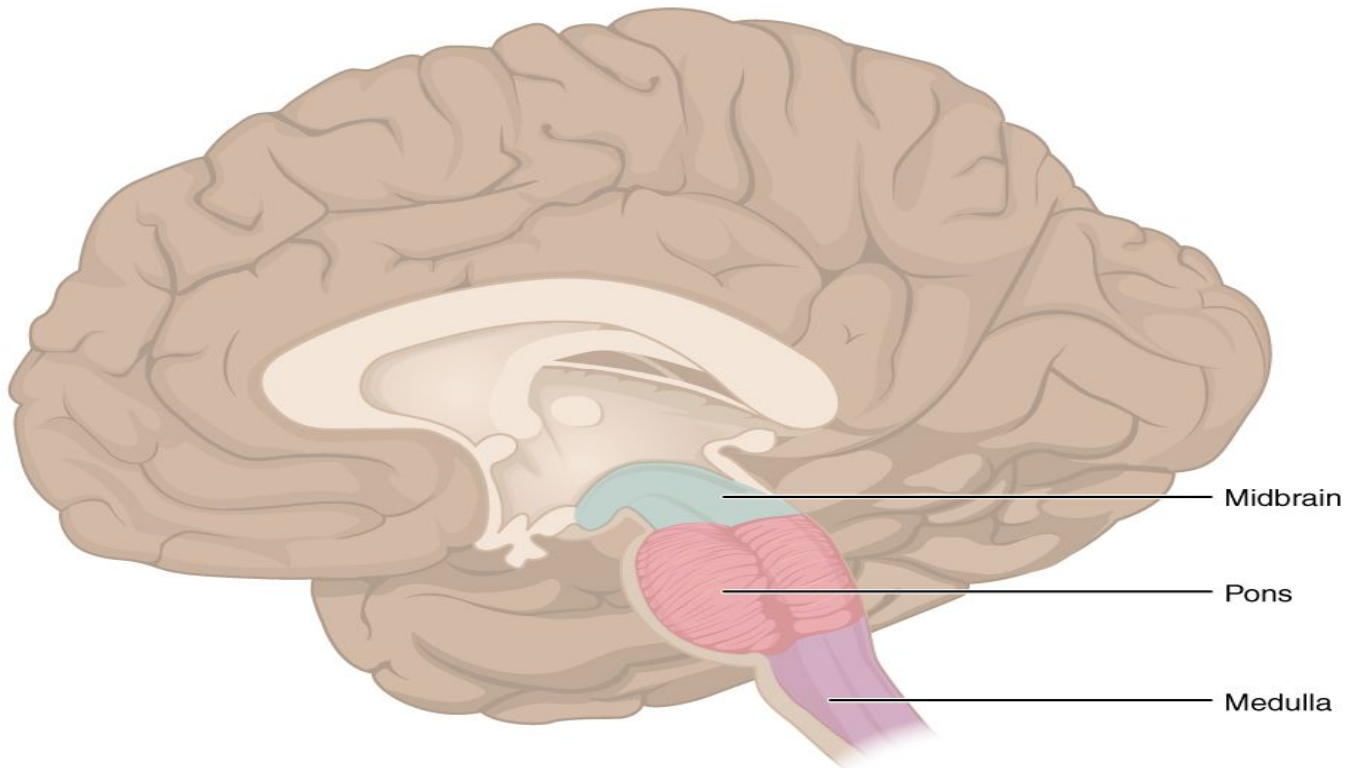
Learning objectives:

At the end of the lecture, the students should be able to describe:

- **Brainstem** and its functions.
- Gross appearance and important internal structures of **medulla oblongata** at different levels.
- Gross appearance and important internal structures of **pons** at different levels.
- Gross appearance and important internal structures of **midbrain** at different levels.

Introduction to Brainstem:

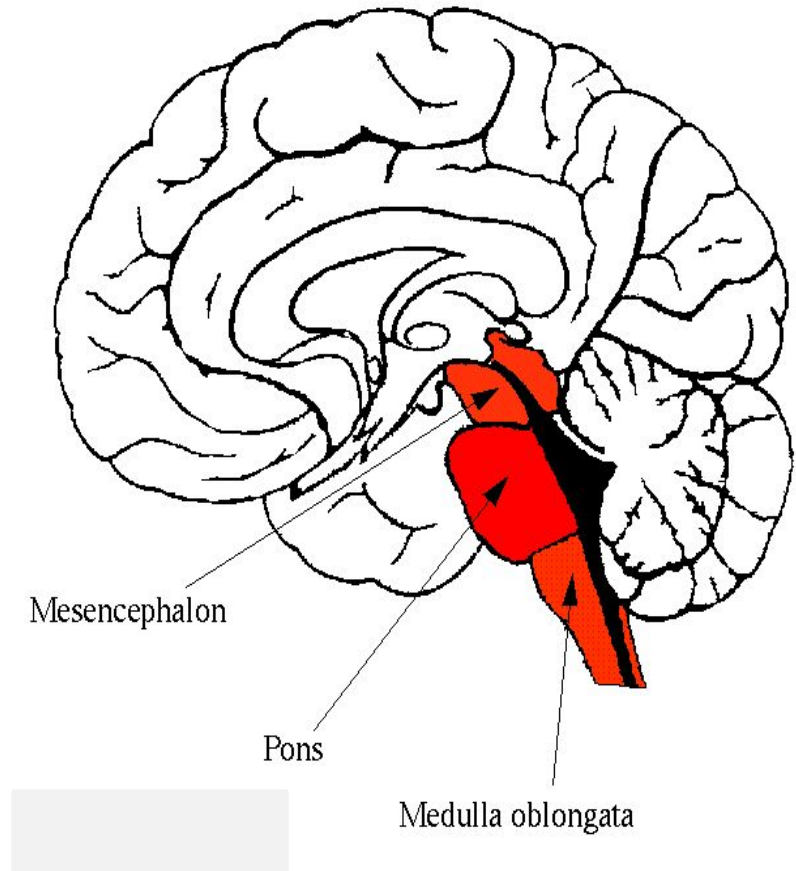
- The brainstem consists of medulla oblongata, pons and midbrain.
- It lies in posterior cranial fossa of skull.
- It has a stalk-like shape which connects spinal cord with forebrain.



Functions of Brainstem:

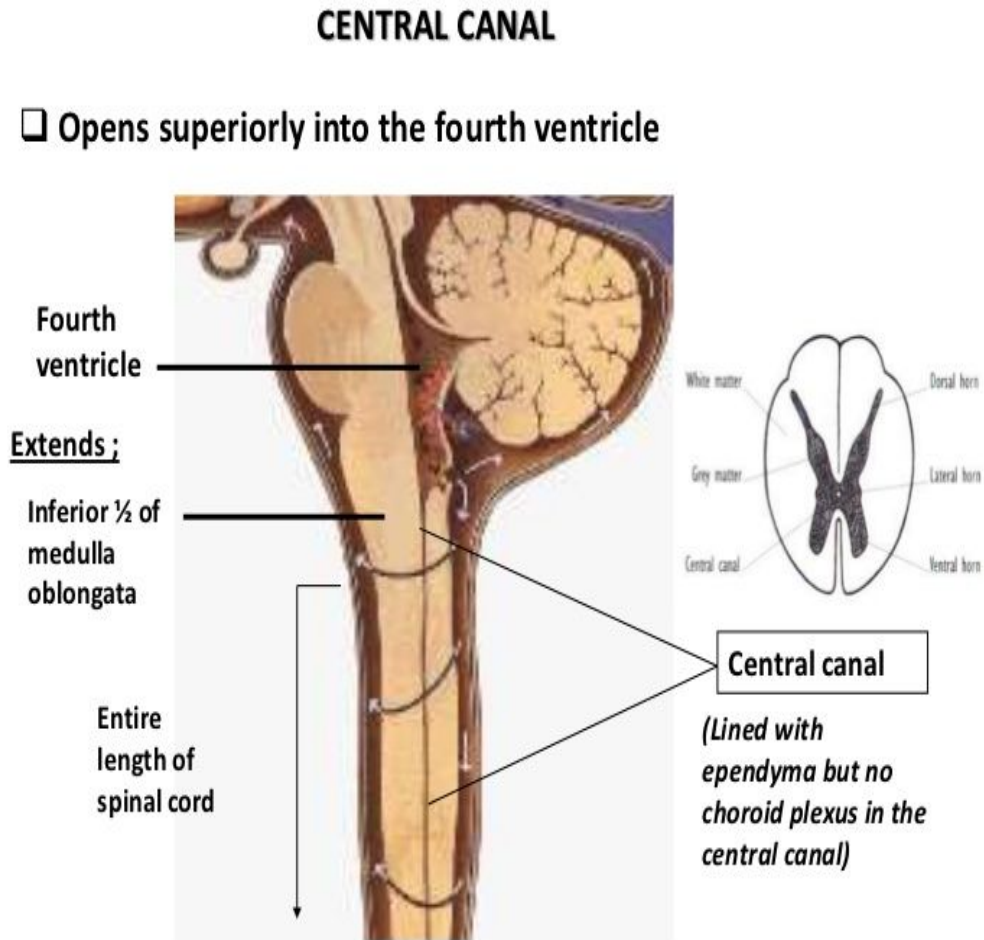
It performs three major functions:

- Serves as a channel for ascending and descending tracts connecting spinal cord to different parts of higher centers of forebrain.
- Contains important reflex centers associated with control of respiration and cardiovascular system and with control of consciousness.
- Contains important nuclei of 3rd to 12th cranial nerves.



The Medulla Oblongata:

- It is continuous superiorly with pons and inferiorly with spinal cord.
- Its junction with spinal cord corresponds to the level of foramen magnum.
- Central canal continues as the same till lower half of medulla but expands in the upper half as cavity of fourth ventricle.



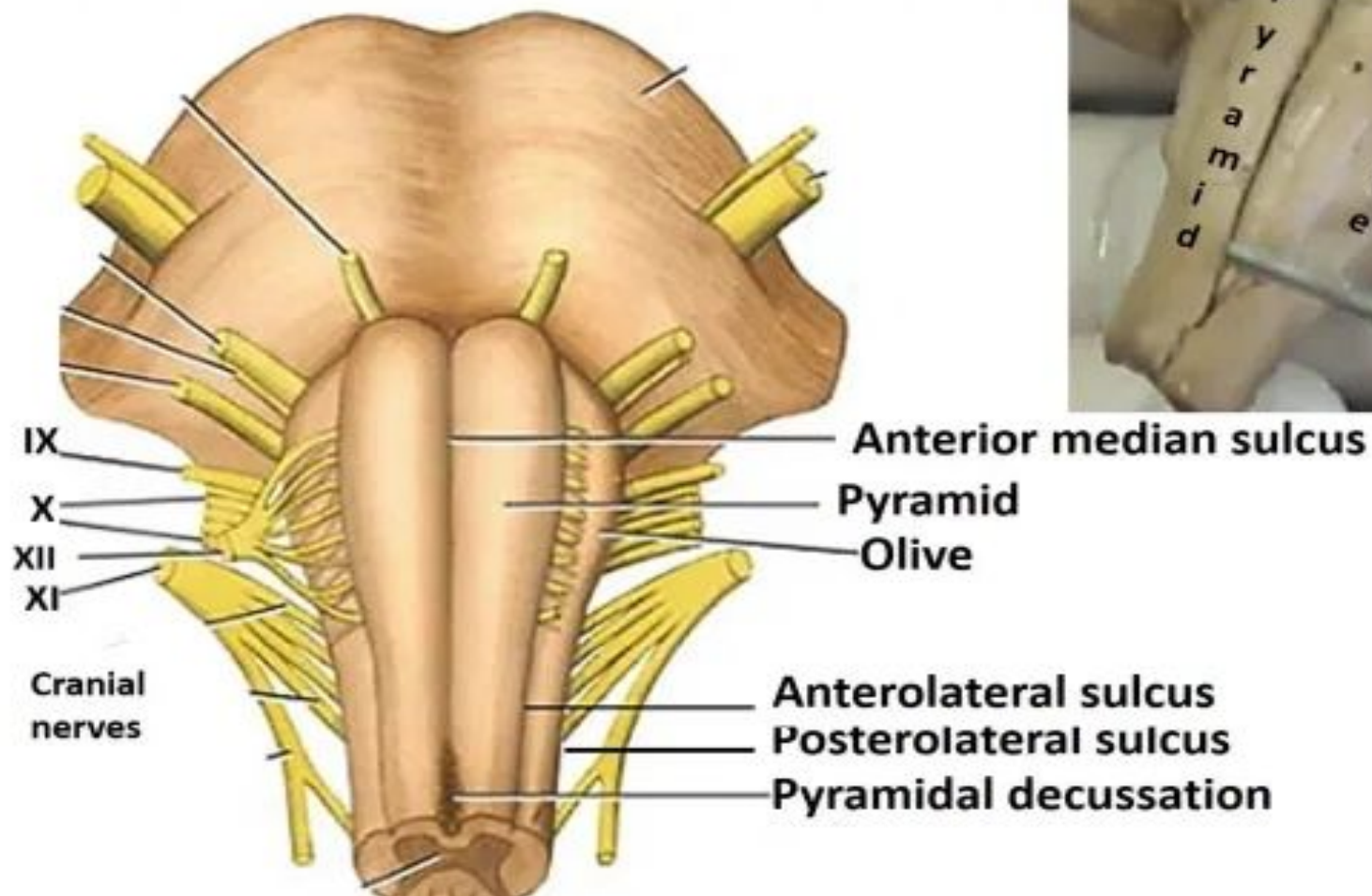
Ventral surface of Medulla:



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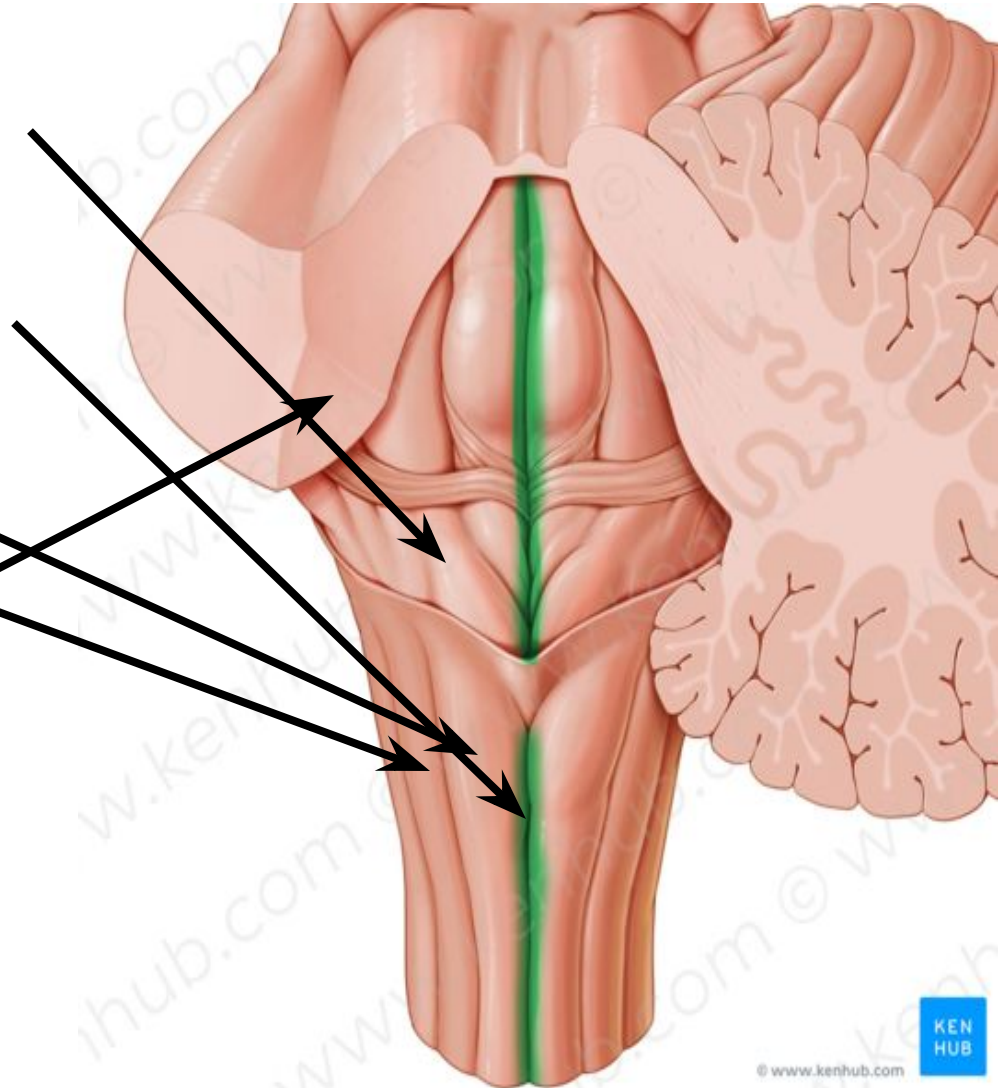
- Anterior median fissure
- Pyramid
- Decussation of pyramids
- Olives
- Anterolateral sulcus
- Roots of hypoglossal (12th) nerve
- Roots of glossopharyngeal (9th), vagus (10th) and cranial roots of accessory (11th) nerve.

External Features: Ventral Surface of Medulla



Dorsal surface of Medulla:

- Floor of fourth ventricle formed by upper half
- Posterior median sulcus
- Gracile tubercle
- Cuneate tubercle
- Inferior cerebellar peduncle



Dorsal surface of medulla

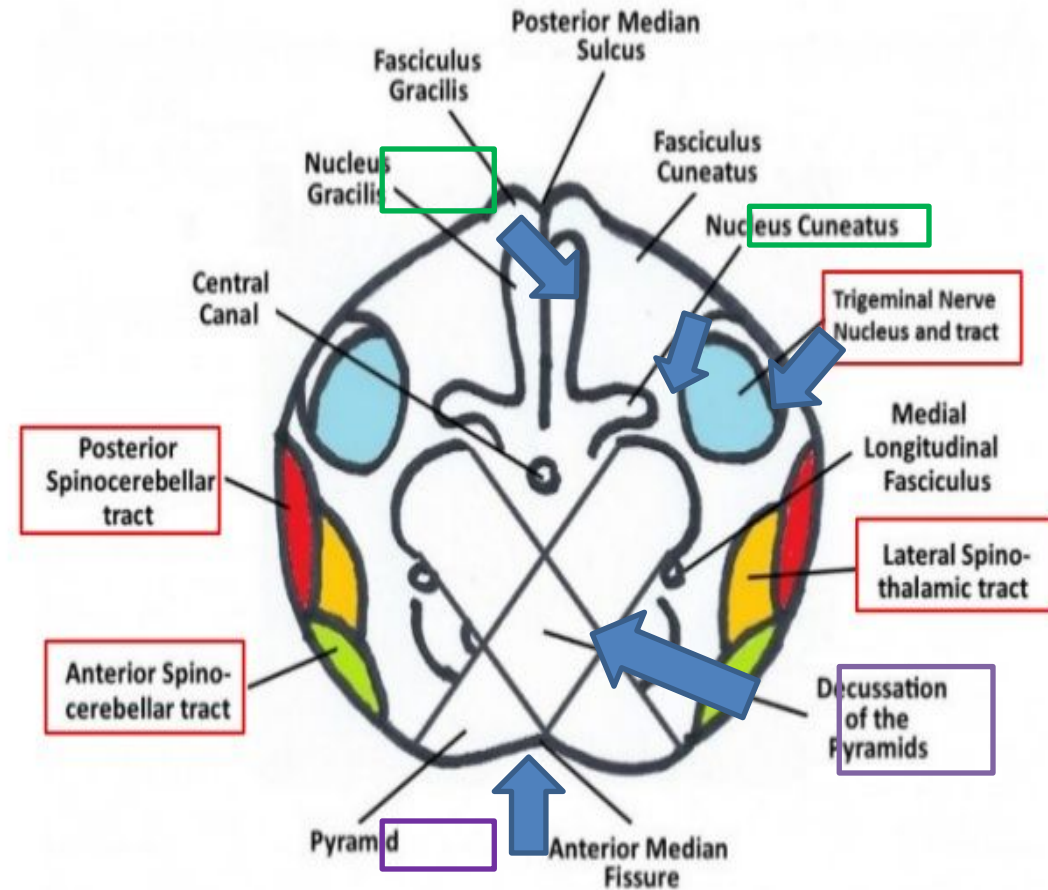


Internal structure of Medulla:

- level of Decussation of Pyramids:

- I. Decussation of pyramids- the **great motor decussation**
- II. Pyramids
- III. Nucleus gracilis & nucleus cuneatus
- IV. Nucleus of spinal tract of trigeminal nerve

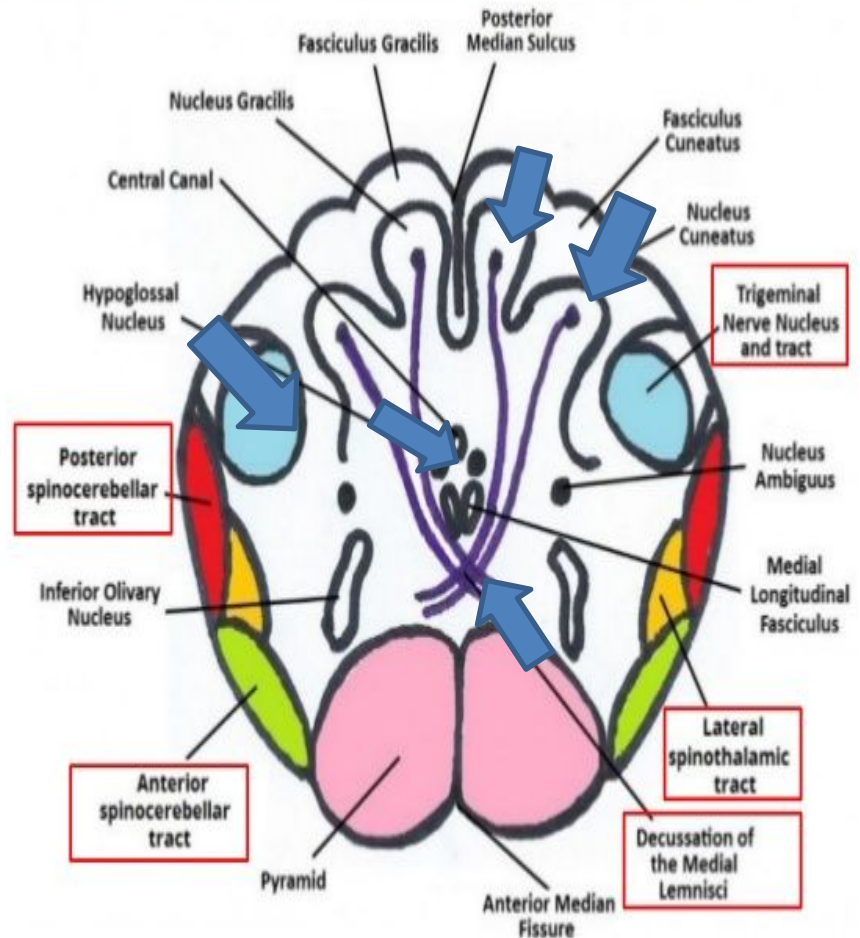
White matter – pyramidal decussation level



- **Level of Decussation of Lemnisci:**

- I. Decussation of lemnisci-the **great sensory decussation**
- II. Nucleus gracilis & nucleus cuneatus
- III. Nucleus of the spinal tract of trigeminal nerve
- IV. Hypoglossal nucleus

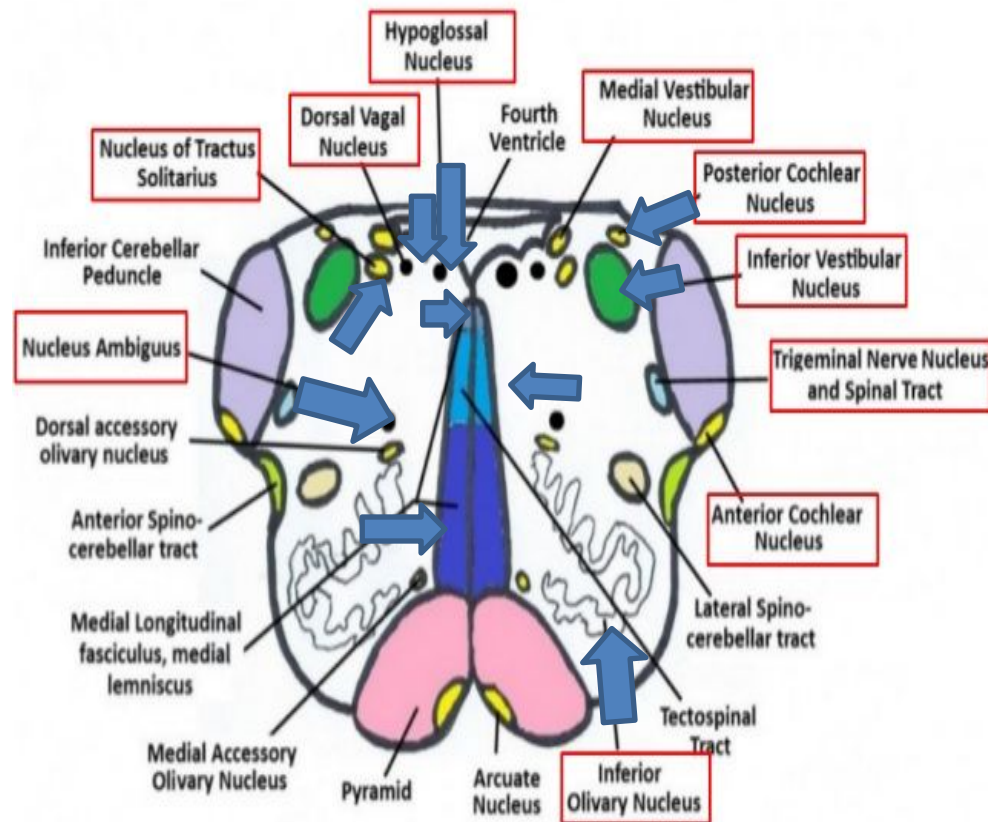
White mater – medial Lemniscus decussation level



- **Level of Olives:**

- I. **Inferior olivary nucleus**
- II. Vestibular nuclei (medial, inferior, lateral & superior)
- III. Cochlear nuclei (anterior & posterior)
- IV. **Nucleus Ambiguus**
- V. Hypoglossal nucleus
- VI. Dorsal vagal nucleus
- VII. **Nucleus of tractus solitarius**
- VIII. Medial lemniscus
- IX. **Tectospinal tract**
- X. Medial longitudinal fasciculus

Open medulla – level of olives



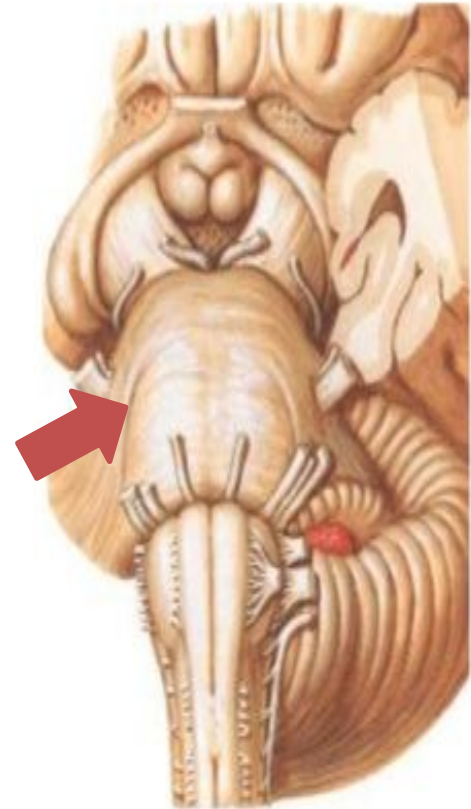
Different levels of Medulla Oblongata showing major structures at each level

Level	Cavity	Nuclei	Motor tracts	Sensory tracts
Decussation of pyramids	Central canal	Nucleus gracilis, nucleus cuneatus, spinal nucleus of trigeminal nerve, accessory nucleus	Decussation of corticospinal tracts, pyramids	Spinal tract of trigeminal nerve, Posterior spinocerebellar tract, lateral spinothalamic tract, anterior spinocerebellar tract
Decussation of leminisci	Central canal	Nucleus gracilis, nucleus cuneatus, spinal nucleus of trigeminal nerve, accessory nucleus, hypoglossal nucleus	Pyramids	Decussation of medial lemnisci, fasciculus gracilis, fasciculus cuneatus, spinal tract of trigeminal nerve, posterior spinocerebellar tract, lateral spinothalamic tract, anterior spinocerebellar tract
Olives	Fourth ventricle	Inferior olivary nucleus, spinal nucleus of trigeminal nerve, vestibular nucleus, glossopharyngeal nucleus, vagal nucleus, hypoglossal nucleus, nucleus ambiguus, nucleus of tractus solitarius	Pyramids	Medial longitudinal fasciculus, tectospinal tract, medial leminiscus, spinal tract of trigeminal nerve, lateral spinothalamic tract, anterior spinocere bellar tract

The Pons:

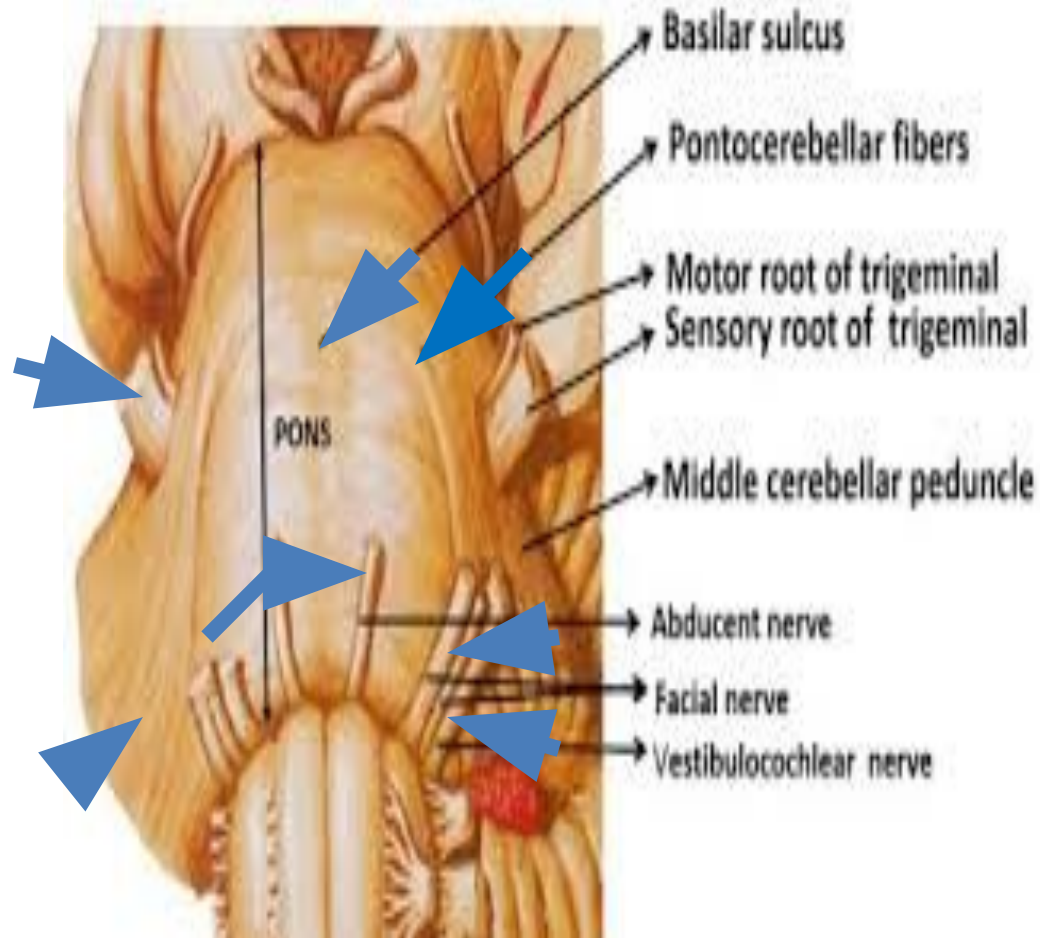
- It lies anterior to cerebellum and connects medulla oblongata to midbrain.
- Its convex anterior surface resembles a bridge connecting the right and left cerebellar hemispheres, thus named as pons.
- Its posterior surface is hidden by cerebellum and forms upper part of fourth ventricle.

PONS



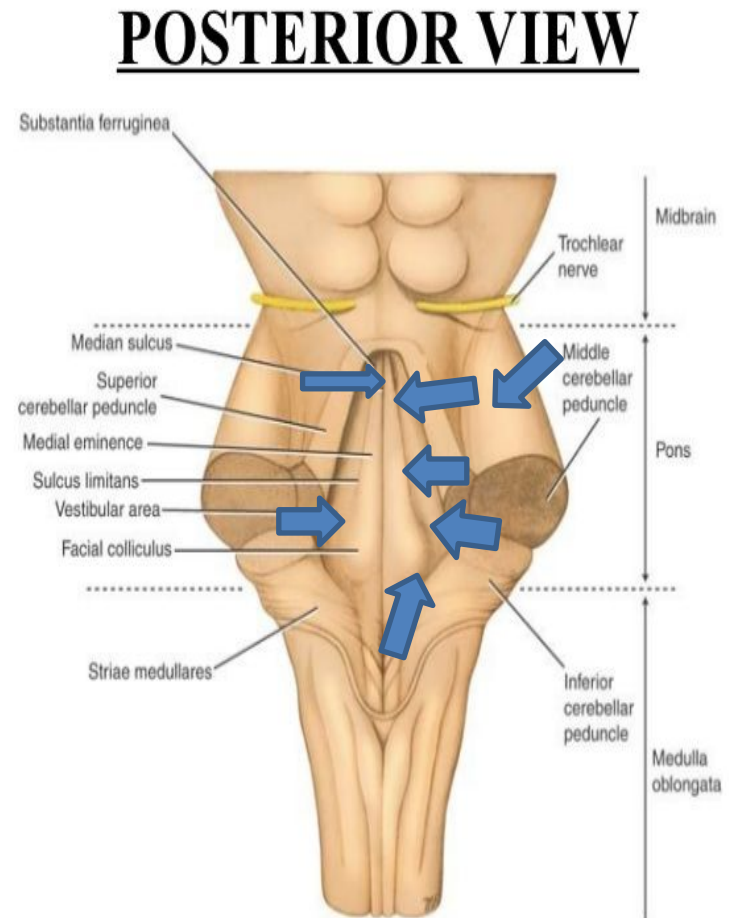
Ventral surface of Pons:

- Transverse pontocerebellar fibers
- Middle cerebellar peduncle
- Basilar groove/sulcus
- Motor & sensory roots of trigeminal nerve
- Nerves arising from ponto-medullary junction from medial to lateral side- abducent, facial & vestibulocochlear nerves



Dorsal surface of Pons:

- I. Middle cerebellar peduncle
- II. Median sulcus
- III. Medial eminence
- IV. Sulcus limitans
- V. Facial colliculus
- VI. Substantia ferruginea
- VII. Area vestibuli

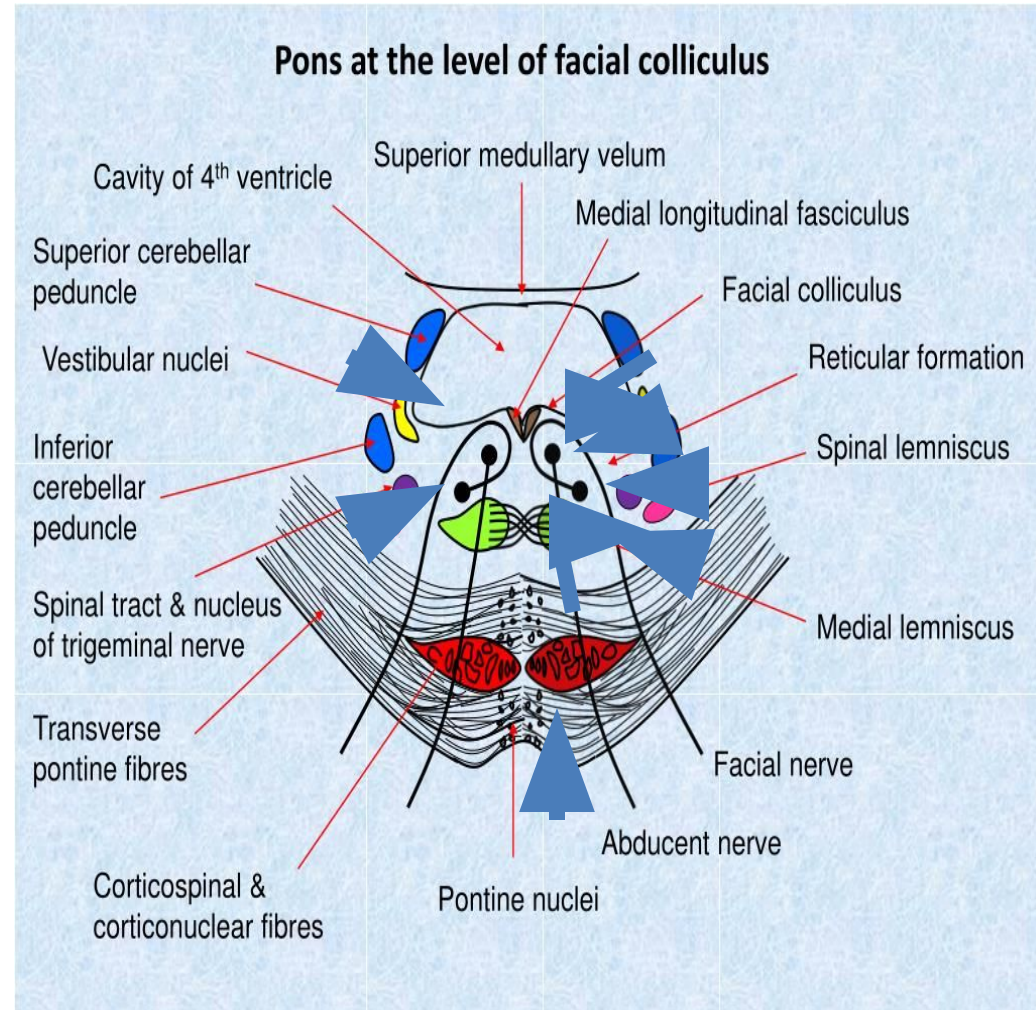


Internal structure of Pons:

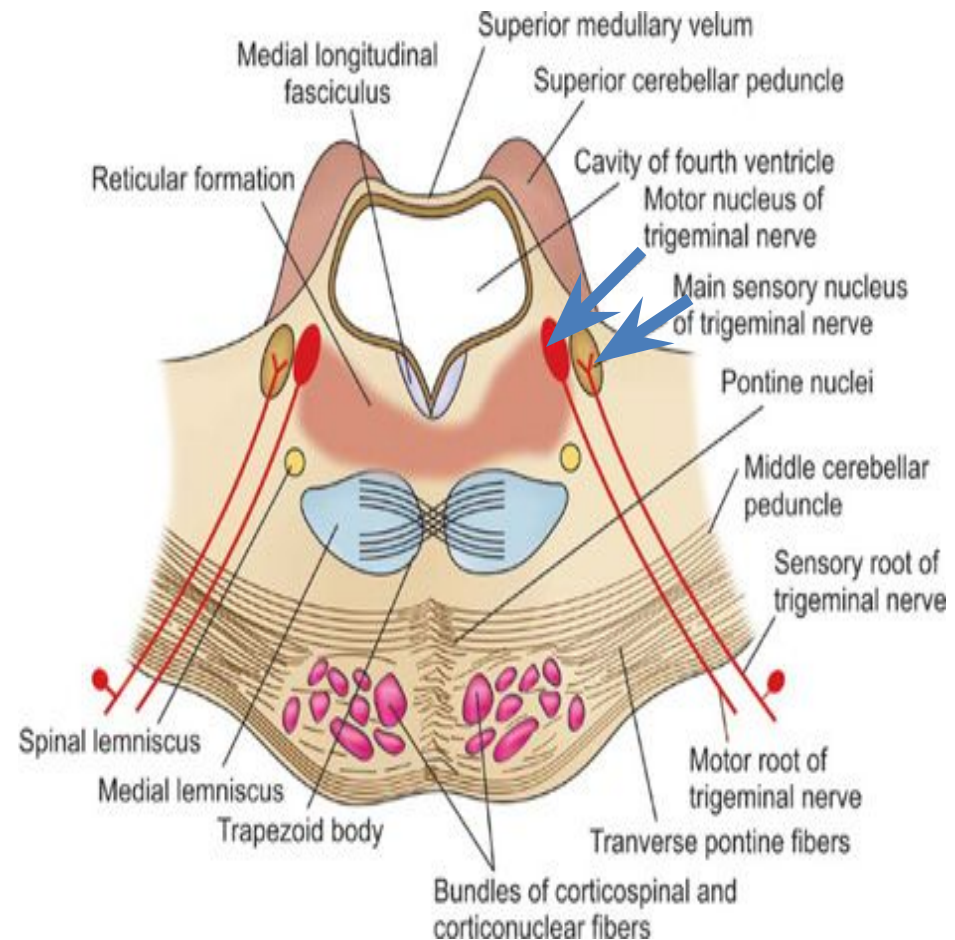
divided into a posterior part, the tegmentum and an anterior basilar part by trapezoid body.

- **Tranverse section through caudal part/facial colliculus:**

- I. Medial lemniscus
- II. Nuclei of abducent & facial nerves
- III. Facial colliculus
- IV. Medial vestibular nucleus
- V. Spinal nucleus of trigeminal nerve
- VI. Trapezoid body
- VII. Pontine nuclei



- **Transverse section through cranial part/level of trigeminal nuclei:**
 - I. Medial lemniscus
 - II. Motor nucleus of trigeminal nerve**
 - III. Principal sensory nucleus of trigeminal nerve**
 - IV. Trapezoid body
 - V. Spinal lemniscus

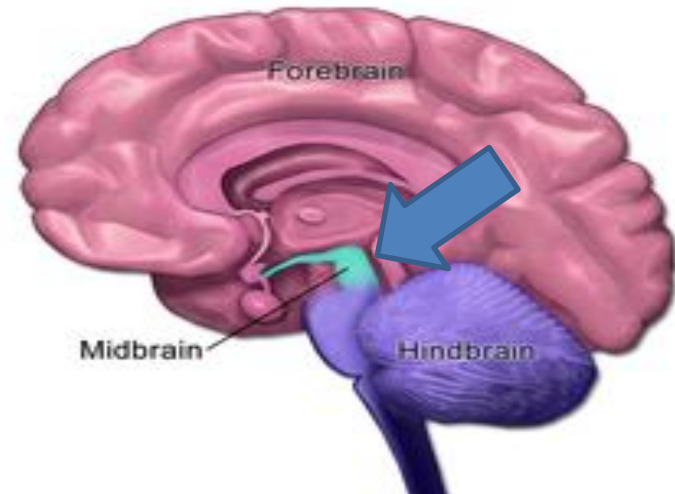
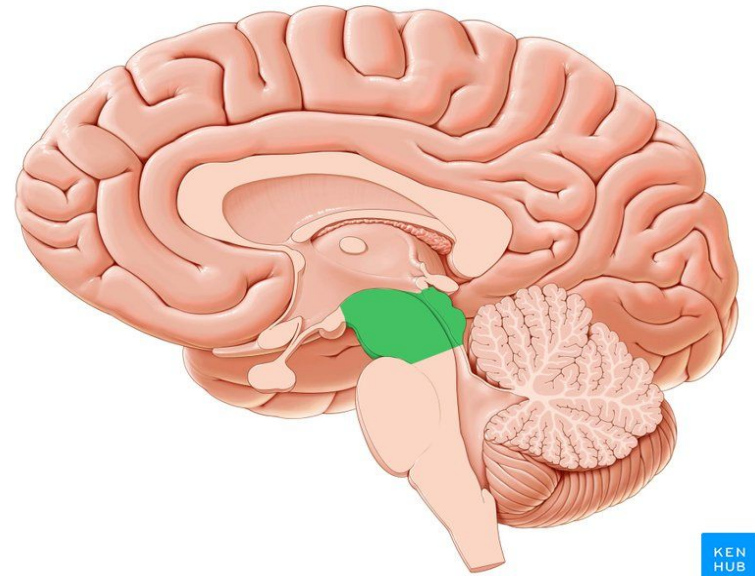


Different levels of Pons showing major structures at each level

Level	Cavity	Nuclei	Motor tracts	Sensory tracts
Facial colliculus	Fourth ventricle	Facial nucleus, abducent nucleus, medial vestibular nucleus, spinal nucleus of trigeminal nerve, pontine nuclei, trapezoid nuclei	Corticospinal & corticonuclear tracts, transverse pontine fibers, medial longitudinal fasciculus	Spinal tract of trigeminal nerve, lateral, spinal and medial lemnisci
Trigeminal nuclei	Fourth ventricle	Main sensory and motor nucleus of trigeminal nerve, pontine nuclei, trapezoid nuclei	Corticospinal & corticonuclear tracts, transverse pontine fibers, medial longitudinal fasciculus	lateral, spinal and medial lemnisci

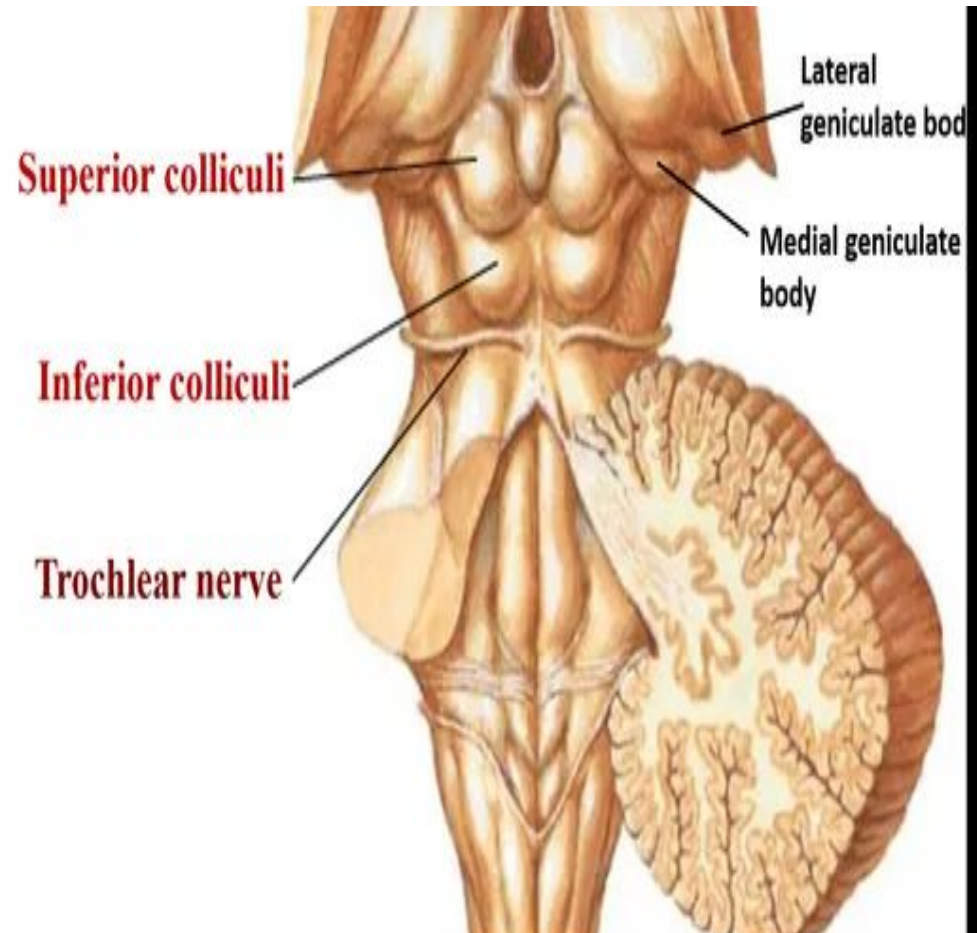
The Midbrain (mesencephalon):

- It connects pons and cerebellum with forebrain.
- It has a narrow cavity, the cerebral aqueduct which is filled with CSF and connects third and fourth ventricles.



Dorsal surface of Midbrain:

- Two pairs of colliculi i.e. **superior colliculi (centers for visual reflexes)** and **inferior colliculi (auditory centers)**.
- Trochlear nerve emerges below inferior colliculi.



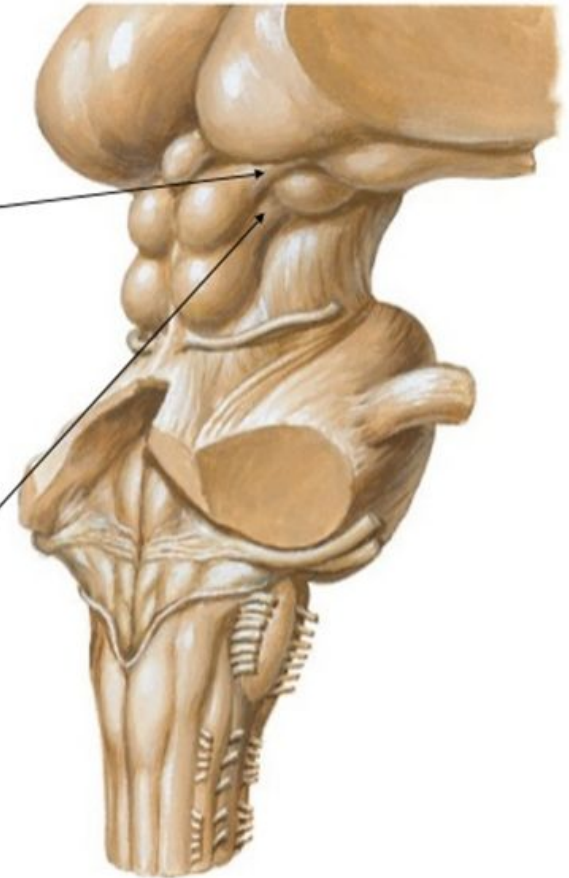
Lateral surface of Midbrain:

- Superior brachium connects superior colliculus to lateral geniculate body and optic tract.
- Inferior brachium connects inferior colliculus to medial geniculate body.

- On the lateral aspect of the midbrain,

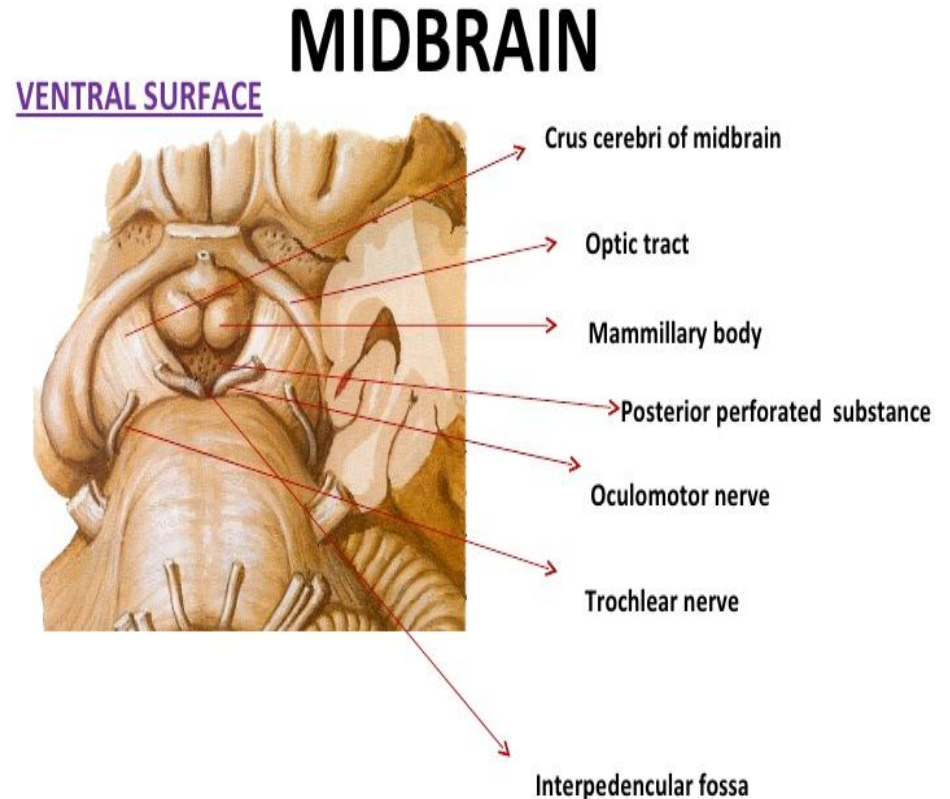
3. **Superior brachium** passes from the superior colliculus to the lateral geniculate body and the optic tract.

4. **Inferior brachium** connects the inferior colliculus to the medial geniculate body.



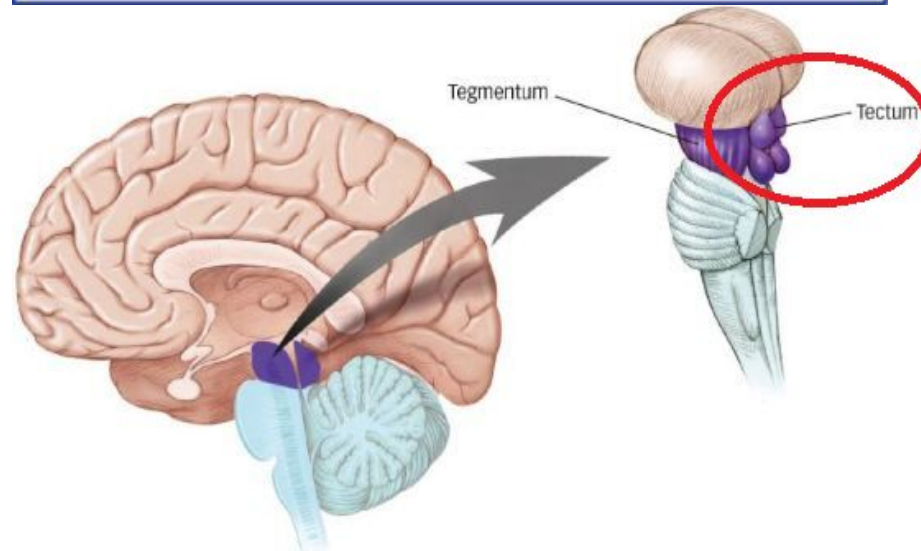
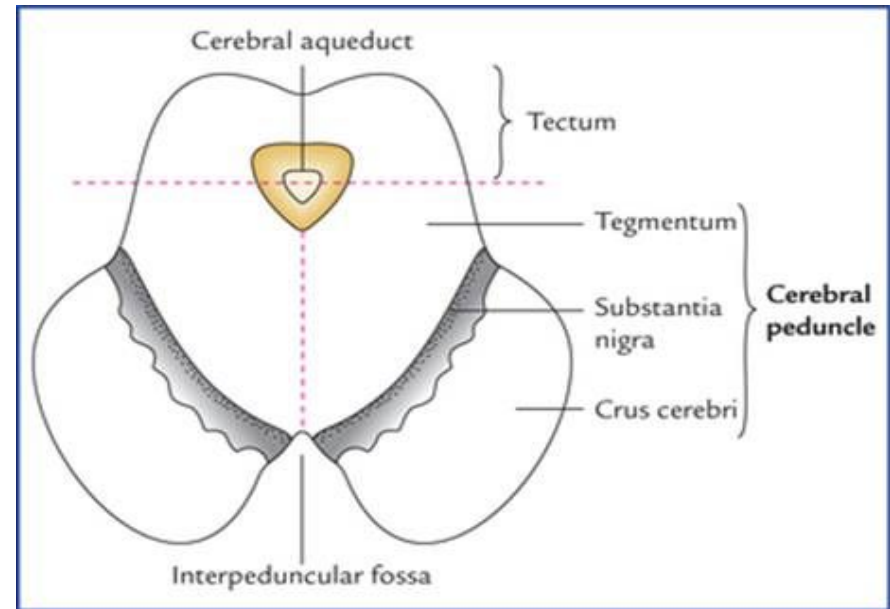
Anterior surface of Midbrain:

- Interpeduncular fossa
- Crus cerebri
- Posterior perforated substance
- Oculomotor nerve



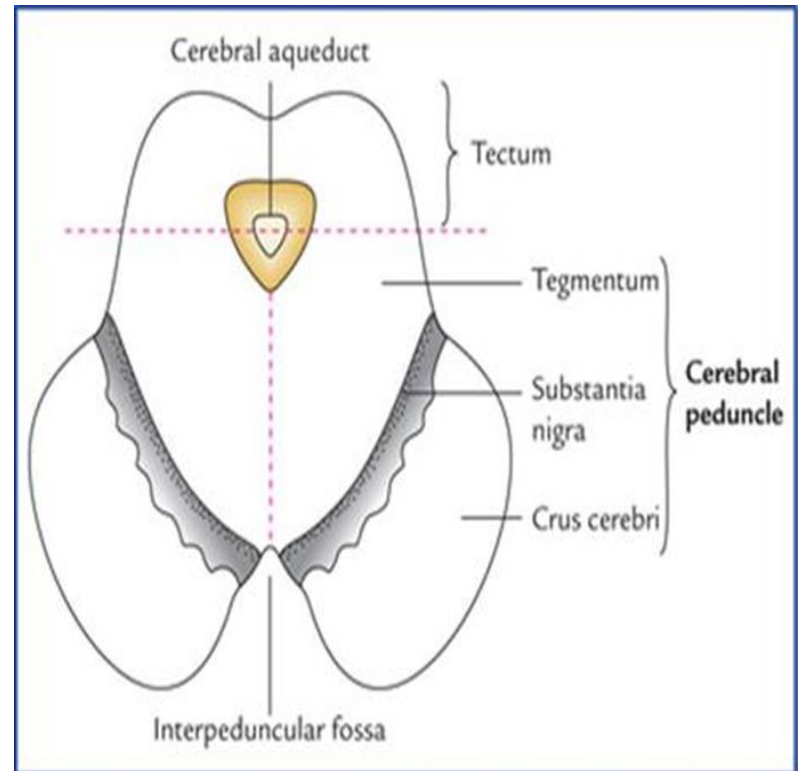
Internal structure of Midbrain:

- Midbrain has two lateral halves, the **cerebral peduncles**.
- Each of these is divided into an **anterior part**, the **crus cerebri**, and a **posterior part**, the **tegmentum** by **substantia nigra**.
- Interpeduncular fossa separates the crura cerebri.
- **Tectum**: is located posterior to cerebral aqueduct and four small swellings, the two superior and two inferior colliculi.

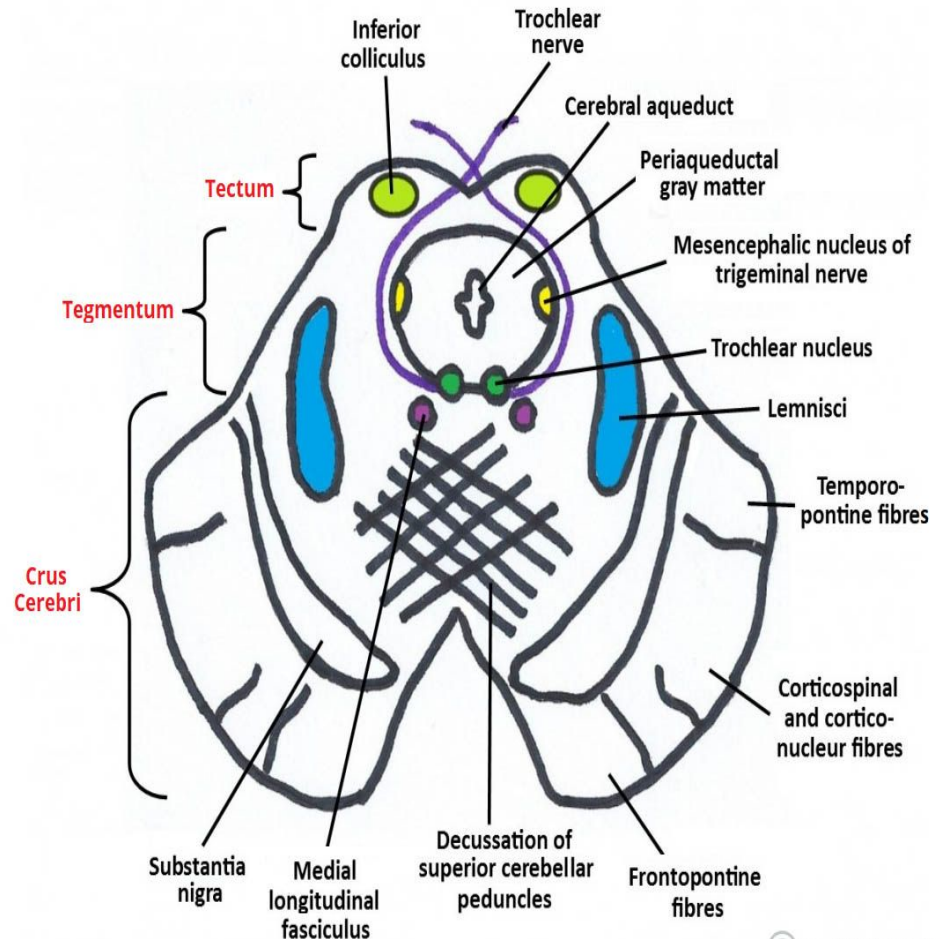


- ❑ **Substantia nigra:** is a large motor nucleus between tectum and crus cerebri in the form pigmented band of gray matter.
 - Has multipolar neurons containing melanin pigment in their cytoplasm.
 - Connected to cerebral cortex, spinal cord, hypothalamus and basal nuclei and is concerned with muscle tone.

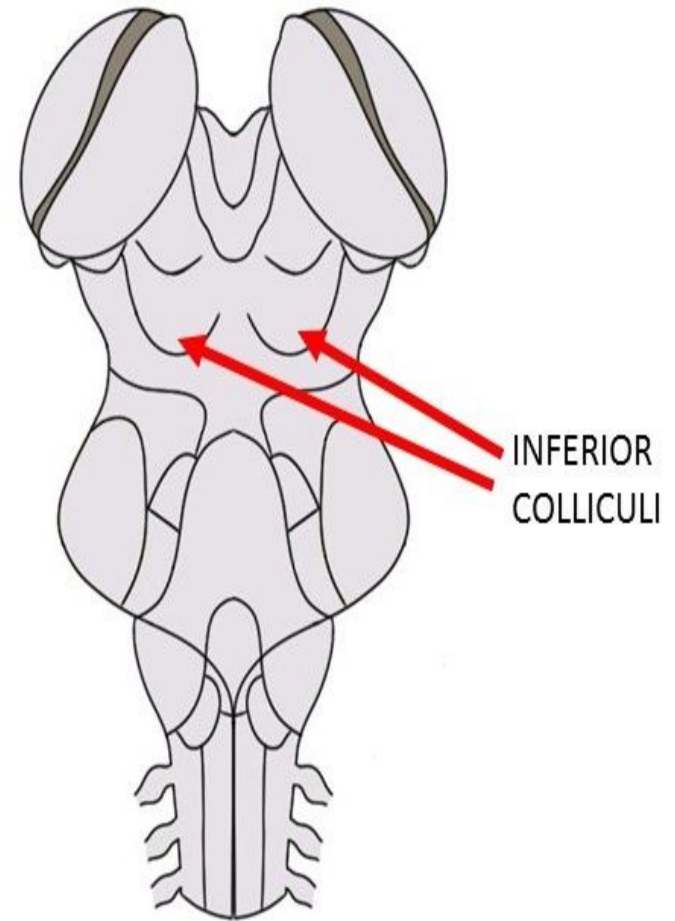
- ❑ **Crus cerebri:** contains important descending tracts.
 - Corticospinal & corticonuclear fibers occupy middle two-third, frontopontine fibers occupy medial and temporopontine fibers occupy lateral part of the crus.
 - These descending tracts connect cerebral cortex to anterior gray column cells of spinal cord, cranial nerve nuclei pons and cerebellum.



- **Transverse section at the level of Inferior Colliculi:**
 - I. Inferior colliculus
 - II. Trochlear nucleus
 - III. Mesencephalic nucleus of trigeminal nerve
 - IV. Decussation of superior cerebellar peduncles
 - V. Substantia nigra

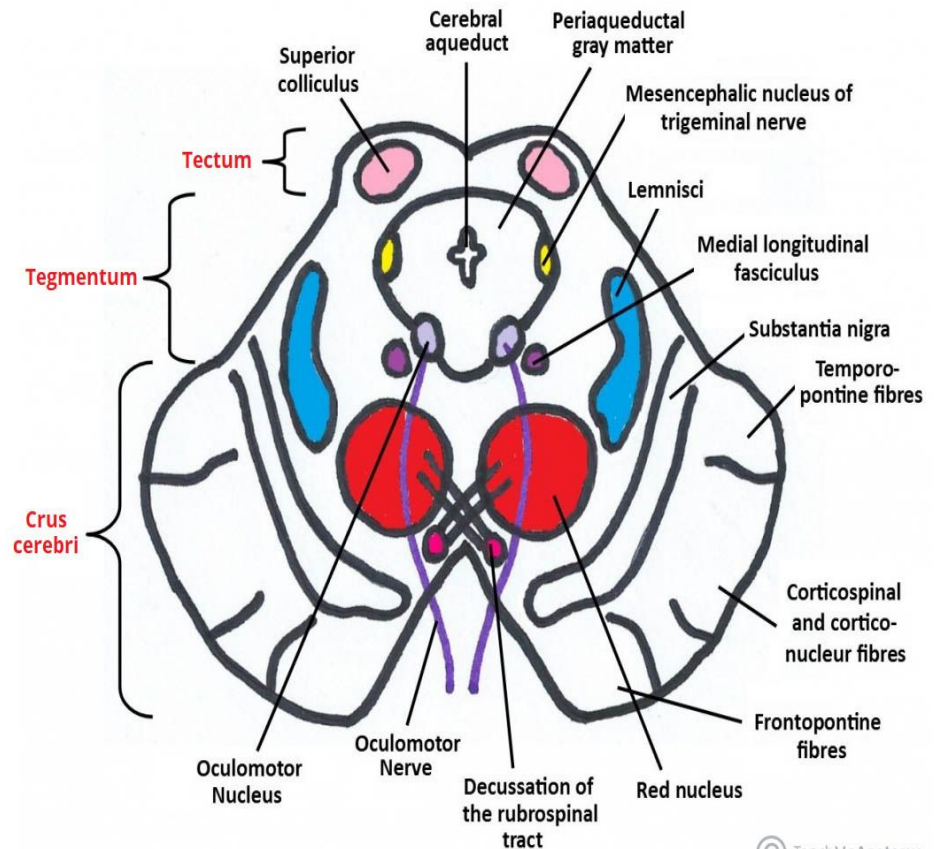


- **Inferior colliculus:** is a large nucleus of gray matter that lies underneath the surface swelling.
- Forms part of auditory pathway.
 - Receives fibers from lateral lemniscus.
 - Is connected to medial geniculate body through inferior brachium.

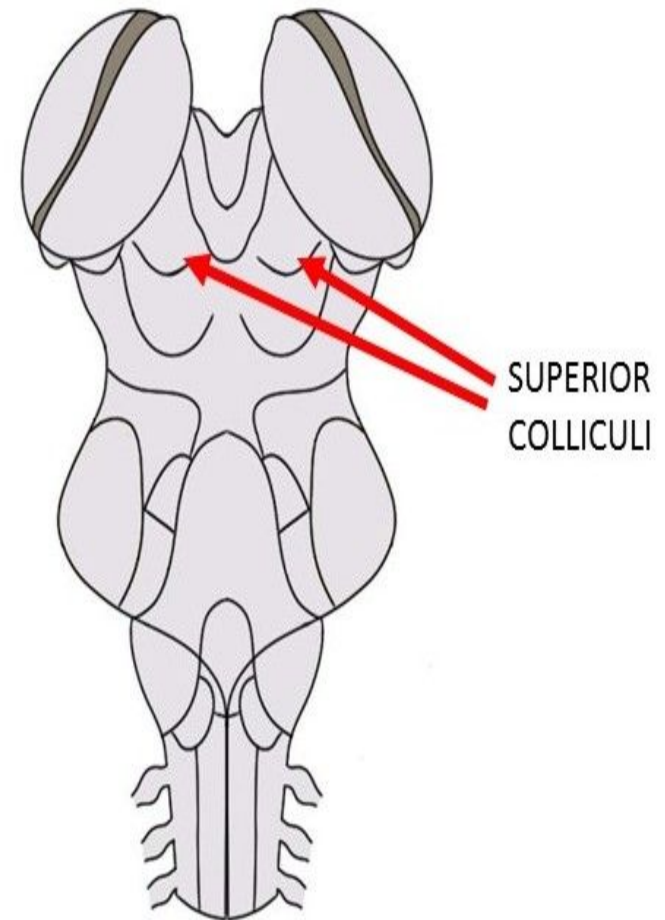


- **Transverse section at the level of Superior Colliculi:**

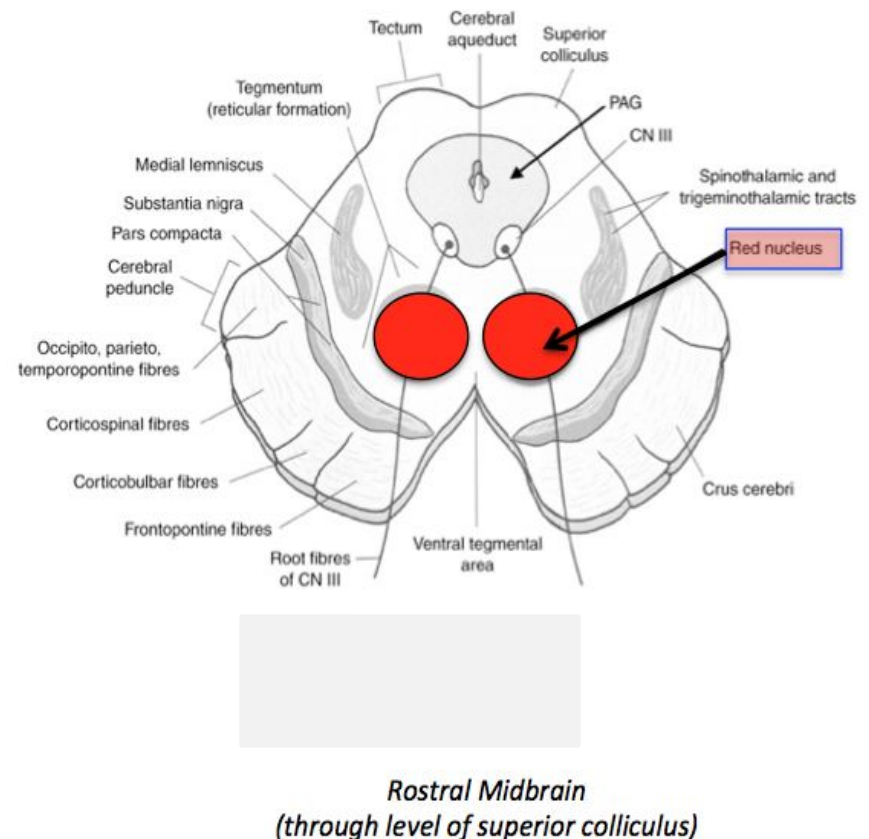
- I. Superior colliculus
- II. Substantia nigra
- III. Mesencephalic nucleus of trigeminal nerve
- IV. Red nucleus
- V. Oculomotor nucleus



- **Superior colliculus:** is a large mass of gray matter situated beneath surface swelling and forms part of visual reflexes.
- Connected to lateral geniculate body through superior brachium.
 - Receives afferent fibers from optic nerve, visual cortex and spinotectal tract.
 - The efferent fibers form tectospinal tract and are responsible for reflex movement of eyes, head & neck in response to visual stimuli.



- **Red Nucleus:** is a spherical mass of gray matter located between cerebral aqueduct and substantia nigra. It appears due to its increased vascularity and the presence of iron-containing pigment in the cytoplasm of its neurons.
- It receives input from opposite cerebellum of opposite side and an input from motor cortex of same side.



Comparison of two levels of midbrain showing main structures at each level

Level	Cavity	Nuclei	Motor tracts	Sensory tracts
Inferior colliculi	Cerebral aqueduct	Inferior colliculus, substantia nigra, trochlear nucleus, mesencephalic nucleus of trigeminal nerve	Corticospinal and corticonuclear tracts, temporopontine, frontopontine, medial longitudinal fasciculus	Lateral, trigeminal, spinal and medial lemnisci, decussation of superior cerebellar peduncles
Superior colliculi	Cerebral aqueduct	Superior colliculus, substantia nigra, oculomotor nucleus, Edinger-Westphal nucleus, red nucleus, mesencephalic nucleus of trigeminal nerve	Corticospinal and corticonuclear tracts, temporopontine, frontopontine, medial longitudinal fasciculus, decussation of rubrospinal tract	Trigeminal, spinal and medial lemnisci,

References:

- **CLINICAL
NEUROANATOMY**

Richard S. Snell

8th edition

Chapter 5

Pages # 196-217